

Queensland Tick Typhus (MONDO:0001118): A Comprehensive Disease-Characteristics Report

Summary

Queensland Tick Typhus (QTT) is an acute, tick-borne **spotted fever group (SFG) rickettsiosis** caused by the obligate intracellular Gram-negative bacterium *Rickettsia australis*. It is transmitted to humans by the bite of hard (ixodid) ticks — principally *Ixodes holocyclus* (the eastern paralysis tick) and *Ixodes tasmani* — along a ~3,200-km strip of eastern coastal Australia, from tropical North Queensland to temperate Tasmania/Victoria. It is a **non-genetic infectious disease**: there are no causal human genes, no inherited susceptibility loci of established clinical importance, and the dominant risk factor is simply environmental/occupational/recreational exposure to tick habitat.

Clinically, QTT is usually a **mild, self-limited eschar-associated febrile illness** characterized by fever, headache, a maculopapular or vesicular rash, an inoculation eschar at the tick-bite site, and regional (draining) lymphadenopathy. However, a clinically significant minority of patients develop **severe disease** requiring intensive-care admission, and death and permanent disability are documented. After tick inoculation the organism first replicates in **dermal/tissue macrophages**, then disseminates to and infects **vascular endothelial cells**, producing the hallmark "**rickettsial vasculitis**" that underlies the rash, eschar, and (in severe cases) increased vascular permeability and multi-organ dysfunction. Host control depends on **innate signaling (TLR4 → MyD88 → ASC inflammasome, dendritic-cell instruction, Th1/IFN- γ)** and, decisively, on **perforin-dependent, MHC class I-restricted CD8⁺ cytotoxic T lymphocytes (CTLs)**, which mouse-model gene-knockout studies show are even more critical to recovery than IFN- γ .

Diagnosis rests on **serology** (indirect immunofluorescence assay [IFA]: a ≥ 4 -fold rise in paired sera or a single IgG titre $\geq 1:64$) combined with **PCR** (real-time qPCR of whole blood, eschar swab, or skin biopsy). First-line treatment is **doxycycline**, which is highly effective and protective against progression to severe disease; **fluoroquinolones should be avoided** because they are associated with worse outcomes in SFG rickettsioses. There is **no licensed**

vaccine; prevention is entirely based on tick-bite avoidance, prompt tick removal, and environmental/personal protective measures. This report synthesizes 11 confirmed findings from 42 reviewed papers into the 15-section disease-characteristics template requested.

Section 1 — Disease Information

Overview. Queensland Tick Typhus is a zoonotic, vector-borne SFG rickettsiosis. It was first recognized clinically in Queensland, and its causative agent, *Rickettsia australis*, was isolated in Queensland in 1950 (type strain "Phillips") — over four decades before comprehensive case reviews consolidated it as a distinct endemic Australian rickettsiosis ([PMID: 22933759](#); [PMID: 1962102](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0001118 (Queensland tick typhus)
Causative organism (NCBI Taxonomy)	<i>Rickettsia australis</i> (species)
MeSH concept	Spotted fever group rickettsiosis / <i>Rickettsia</i> infections (no unique QTT MeSH descriptor)
ICD-10	A77.3 (Spotted fever due to <i>Rickettsia australis</i>)
ICD-11	1C30.2 / spotted fever group rickettsiosis category
OMIM / Orphanet	Not applicable (infectious, non-Mendelian; no OMIM entry)

Synonyms / alternative names. Queensland tick typhus; Australian tick typhus; North Queensland tick typhus; *Rickettsia australis* infection; Australian spotted fever (in part — note that "Australian spotted fever" also encompasses Flinders Island spotted fever caused by *R. honei*, a distinct co-endemic agent).

Source of information. The knowledge base for QTT is derived from **aggregated disease-level resources** — published case series, hospital audits, serosurveys, microbiological/genomic characterizations, and animal-model immunology — rather than from individual-patient EHR data.

Section 2 — Etiology

Primary cause (infectious). QTT is caused by infection with *Rickettsia australis*, an obligately intracellular SFG rickettsia, delivered into the skin during the blood meal of an infected ixodid tick (PMID: 1962102). It is **not** a genetic, autoimmune, metabolic, or neoplastic disease.

Risk factors. - **Genetic risk factors (human):** None of established clinical relevance. No causal variants, susceptibility loci, or modifier genes have been identified for human QTT. Innate-immunity genes (*MYD88*, *TLR4*, inflammasome components, MHC-I, perforin/*PRF1*) are mechanistically important in **animal models** (see Sections 4 and 6) but are not validated human susceptibility markers. - **Environmental / behavioral risk factors:** Exposure to tick-infested vegetation in endemic eastern-coastal Australia is the principal risk factor — including bushwalking, gardening, camping, and outdoor occupations. Epidemiology shows a skew toward **older males** and a **late spring/early summer (November–December)** seasonal peak, reflecting exposure patterns and tick questing behavior (PMID: 31268225).

Protective factors. No genetic protective variants are described in humans. The single most effective environmental protective factor is **tick-bite avoidance** (protective clothing, repellents, tick checks, prompt removal). Prompt **doxycycline** after infection is protective against severe disease progression (PMID: 21642652).

Gene–environment interactions. Not characterized in humans. The relevant "gene–environment" axis is between the pathogen's virulence factors and the host's innate/adaptive immune genotype, demonstrated experimentally in mice (*MyD88*, *TLR4*/ASC, perforin, MHC-I).

Section 3 — Phenotypes

QTT produces a stereotyped SFG-rickettsiosis phenotype. Sexton et al. summarized the classic presentation: "*QTT is usually a mild disease. Patients often have regional lymphadenopathy and eschars. Some have vesicular rashes.*" (PMID: 1962102).

Phenotype	Type	HPO suggestion	Typical frequency / notes
Fever	Symptom/sign	HP:0001945 (Fever)	Near-universal; commonly reported (PMID: 31268225)
Inoculation eschar (tache noire)	Clinical sign	HP:0200041 (Skin ulcer) / eschar	Frequent; hallmark at tick-bite site (PMID: 1962102)
Maculopapular / vesicular rash	Physical manifestation	HP:0000988 (Skin rash); HP:0200037 (Vesicle)	Common; vesicular rash a distinctive QTT feature
Regional lymphadenopathy	Clinical sign	HP:0002716 (Lymphadenopathy)	Common (draining nodes)
Headache	Symptom	HP:0002315 (Headache)	Common
Lethargy / malaise	Symptom	HP:0001254 (Lethargy)	Common (PMID: 31268225)
Myalgia	Symptom	HP:0003326 (Myalgia)	Common
Elevated transaminases	Lab abnormality	HP:0002910 (Elevated hepatic transaminase)	Frequent in SFG rickettsioses
Thrombocytopenia	Lab abnormality	HP:0001873 (Thrombocytopenia)	Variable; marks more severe disease
Severe disease (ICU / organ dysfunction)	Clinical course	critical illness	~13–22% ICU in SFG hospital cohorts (PMID: 31318873; PMID: 32959771)

Onset / severity / progression. Adult-onset predominant (any age exposed can be affected). Severity is **variable** — mild in most, but severe/fatal in a minority. Course is **acute and self-limited** with appropriate therapy. In a 20-year North Queensland audit of 135 cases (95 scrub typhus, 37 SFG/QTT), 18/135 (13%) required ICU; the SFG subgroup had a **higher ICU rate (8/37, 22%)** and 3/37 (8%) had severe disease — “1 died, 2 developed permanent disability” — versus 0/95 scrub typhus ($p = 0.02$) (PMID: 31318873). By contrast, paediatric rickettsial infection in tropical Australia has “a relatively benign clinical course” with no ICU admission or death among 15 children (PMID: 32252063).

Quality-of-life impact. Not formally measured with instruments (EQ-5D/SF-36) for QTT. For most patients the illness is acute and fully reversible; the QoL burden concentrates in the severe subset who suffer prolonged hospitalization, ICU stay, or (rarely) permanent disability.

Section 4 — Genetic/Molecular Information

Human genetics. QTT is an infectious disease with **no causal human genes, no pathogenic germline/somatic variants, no modifier genes, no epigenetic disease signatures, and no chromosomal abnormalities**. Sections requesting ClinVar/gnomAD/COSMIC variant data are not applicable.

Pathogen genome (the relevant molecular biology). The *R. australis* strain Phillips genome is a compact **1.29-Mb** chromosome accompanied by **two plasmids**, and is "*highly similar to that of Rickettsia akari*" (PMID: 22933759). Phylogenetically, multiple independent gene analyses place *R. australis* firmly within the "**Rickettsia akari group**" of the SFG, alongside *R. akari* (rickettsialpox agent) and the ELB/*R. felis* agent (gene D: PMID: 11491333; citrate synthase *gltA*: PMID: 9103608). Notably, *R. australis* is described as "*the most divergent rickettsia of the spotted fever group*" on the basis of its outer-membrane protein A and B genes (PMID: 11034486). Key rickettsial virulence/antigen loci include *ompA*, *ompB* (surface cell antigens mediating adhesion/invasion), the 17-kDa antigen gene, and *gltA*.

Host immune-gene involvement (model organism). Genes shown experimentally to govern outcome are *Myd88*, *Tlr4*, inflammasome adaptor *Pycard*/ASC, MHC class I, and perforin (*Prf1*) — see Section 6.

Section 5 — Environmental Information

Environmental factors. The disease is defined by an environmental exposure: contact with tick habitat in the humid sclerophyll forests, coastal scrub, and peri-urban bushland of eastern Australia. No chemical toxin, radiation, or pollutant is involved.

Lifestyle factors. Outdoor recreation and occupation (bushwalking, camping, gardening, forestry, field work) increase exposure. Seasonality (late spring–summer) reflects tick activity (PMID: 31268225).

Infectious agent. *Rickettsia australis* — an obligately intracellular, cytosolically replicating, Gram-negative alphaproteobacterium (order Rickettsiales, SFG, *R. akari* group). It is a pathogen, not a chemical entity. Vectors *Ixodes holocyclus* and *Ixodes tasmani* maintain the organism; small native mammals (rodents, bandicoots, marsupials) serve as reservoir/amplifying hosts, with humans as **accidental dead-end hosts** (PMID: 1962102; PMID: 25434042).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. An **infected ixodid tick** (*I. holocyclus* / *I. tasmani*) attaches and takes a blood meal, **inoculating** *R. australis* into the dermis at the bite site. → (demonstrated)
2. Local infection and replication in **dermal/tissue macrophages** — "*Macrophages are one of the initial targets for rickettsiae after inoculation by ticks*" — **leads to** an early intracellular replication niche and a localized inflammatory/necrotic focus that becomes the **eschar** (PMID: 30297526). → (demonstrated in mouse model)
3. Rickettsiae disseminate (lymphatics → draining nodes, producing **regional lymphadenopathy**; and haematogenously) and **infect vascular endothelial cells**, the pathogen's principal target — "*the pathogen's affinity for endothelium lining the blood vessels*" — **resulting in** endothelial infection throughout small vessels (PMID: 19327117). → (demonstrated)
4. Endothelial infection **causes** "*vascular inflammation, insult to vascular integrity and compromised vascular permeability, collectively termed 'Rickettsial vasculitis'*" (PMID: 19327117). → (demonstrated)
5. Rickettsial vasculitis **produces** the clinical phenotype: perivascular mononuclear infiltrate → **maculopapular/vesicular rash**; microvascular injury → increased permeability. → (inferred from SFG pathology, consistent across the group)
6. **Branch (host innate immunity):** Pathogen-associated molecular patterns engage **TLR4** → **MyD88** signaling and activate the **ASC inflammasome**, driving dendritic-cell instruction, macrophage/neutrophil recruitment, and a **Th1 cytokine milieu** (IFN- γ , IL-6, IL-1 β , IL-12p40) that **restrains** bacterial burden (PMID: 32014896; PMID: 26755162). → (demonstrated in mouse model)

7. **Branch (host adaptive immunity, decisive): MHC class I-restricted, perforin-dependent CD8⁺ CTLs** recognize and kill infected macrophages and endothelial cells, **clearing** the infection — this effector arm is *"more critical to recovery... than were the effects of IFN-gamma"* (PMID: 11179362). → (demonstrated in mouse model)
8. **Outcome branch:** When innate + CD8 CTL immunity (aided by timely doxycycline) controls the organism → **mild, self-limited illness**. When bacterial dissemination outpaces immunity (older age, delayed treatment) → widespread endothelial injury, capillary leak, organ dysfunction → **severe disease / death** (PMID: 31318873). → (inferred, supported by clinical severity data)

Detail by category

- **Molecular pathways:** TLR4→MyD88→NF-κB innate signaling; ASC/NLR inflammasome → caspase-1 → IL-1β maturation; Th1/IFN-γ-STAT1 axis; MHC-I antigen-presentation and perforin/granzyme cytotoxicity pathway.
- **Cellular processes:** Intracellular bacterial invasion and cytosolic replication; inflammation; macrophage and endothelial infection; CTL-mediated cytotoxicity of infected cells; endothelial activation and increased permeability.
- **Protein dysfunction:** Not a protein-misfolding disease; pathology is driven by pathogen invasins (OmpA/OmpB) and host endothelial-barrier dysfunction.
- **Immune involvement:** Central and protective (innate TLR4/MyD88/ASC; adaptive CD8 CTL). Loss of MyD88 causes *"severely impaired bacterial clearance in vivo"* (PMID: 26755162).
- **Tissue-damage mechanism:** Rickettsial vasculitis — immune-mediated and infection-mediated microvascular injury with increased permeability (PMID: 19327117).

Suggested ontology terms. GO: GO:0002224 (toll-like receptor signaling pathway), GO:0045087 (innate immune response), GO:0001913 (T cell-mediated cytotoxicity), GO:0006954 (inflammatory response), GO:0050830 (defense response to Gram-negative bacterium). CL: CL:0000235 (macrophage), CL:0000071 (blood vessel endothelial cell), CL:0000625 (CD8-positive, alpha-beta T cell), CL:0000451 (dendritic cell).

Section 7 — Anatomical Structures Affected

- **Primary target:** systemic **small-vessel vascular endothelium** (UBERON:0001986 endothelium; UBERON:0000115 endothelial layer of blood vessel) — the unifying lesion of SFG rickettsiosis ([PMID: 19327117](#)).
 - **Skin (UBERON:0002097):** eschar at inoculation site; maculopapular/vesicular rash.
 - **Lymphatic system (UBERON:0000465):** regional draining **lymph nodes** (UBERON:0000029) — lymphadenopathy.
 - **Body systems involved:** cardiovascular (microvasculature), integumentary, lymphoid/immune; in severe disease, potential secondary involvement of lung, liver, kidney, and CNS via generalized vasculitis/capillary leak.
 - **Cell populations targeted:** vascular endothelial cells (CL:0000071) and macrophages (CL:0000235); CD8⁺ CTLs (CL:0000625) are the key effector population.
 - **Subcellular level:** *R. australis* is a **cytosolic** pathogen — it replicates free in the host-cell cytoplasm (GO:0005829 cytosol), escaping the phagosome.
 - **Lateralization:** the eschar/lymphadenopathy are **localized/unilateral** to the bite site; the rash and vasculitis are **generalized/bilateral**.
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Section 8 — Temporal Development

- **Onset: acute**, following an incubation period of roughly 7–10 days after the tick bite; patients in the ICU series presented a median of 7 (IQR 5–10) days after symptom onset ([PMID: 32959771](#)).
- **Progression / course: self-limited** in most, resolving over days to ~1–2 weeks with therapy; **not** chronic, relapsing, or lifelong. A minority progress to severe multi-organ disease.
- **Treatment-response kinetics:** In hospitalized scrub typhus/QTT patients on anti-rickettsial therapy, 32/58 (56%) **had delayed defervescence (>48 h)**; delayed defervescence was associated with older age (median 52 vs 40 y, $p = 0.05$), higher ICU admission (38% vs 12%, $p = 0.02$), and longer stay — challenging the dogma that persistent fever >48 h implies a wrong diagnosis ([PMID: 33075531](#)).
- **Critical period for intervention:** early — doxycycline given "*prior to deterioration of disease... protected patients from development of severe*" disease in the analogous Mediterranean spotted fever cohort ([PMID: 21642652](#)).

- **Remission:** treatment-induced (doxycycline) and, in mild cases, potentially spontaneous.

Section 9 — Inheritance and Population

Inheritance. Not applicable — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier-frequency considerations.

Epidemiology. - QTT/Australian SFG rickettsiosis is a **rare, notifiable, seasonally patterned** infection. In Tasmania (2012–2017), *"The mean number of cases per year was 3.0 (population rate 0.6 per 100,000 population/year); 60% of cases occurred in November and December. Cases were more commonly older males"* (PMID: 31268225). (Note: Tasmanian SFG cases are predominantly Flinders Island spotted fever caused by *R. honei*, a co-endemic agent; true QTT/*R. australis* predominates on the mainland eastern seaboard.) - Ascertainment/incidence is **rising**: in a North Queensland tertiary hospital, *"There were nine hospitalizations during the first 5 years of the study period and 81 in the last 5 years (p for trend = 0.003)"* (PMID: 31318873).

Population demographics. - **Geographic distribution:** eastern coastal Australia across a **~3,200-km span** *"from tropical to temperate climates"* — Queensland, New South Wales, Victoria, and Tasmania (PMID: 1962102). - **Sex / age:** skewed to **older males** (PMID: 31268225); paediatric cases occur but run a milder course (PMID: 32252063).

Section 10 — Diagnostics

Clinical/laboratory tests. - **Serology (mainstay):** indirect immunofluorescence assay (IFA); diagnosis by a **≥4-fold rise** in IgG between acute and convalescent sera, or a single **IgG titre ≥1:64** with a compatible illness (PMID: 1962102; titre convention as applied in SFG serosurveys, PMID: 31288833). Limitation: antibodies are absent early in illness. - **Molecular (PCR):** real-time qPCR targeting SFG/*R. australis*-specific genes (e.g., *gltA*, *ompA*, *ompB*, 17-kDa) on **whole blood, eschar swab, or skin biopsy**; validated *R. australis* real-time PCR assays are in routine reference use (PMID: 22092999). - **Recommended strategy:** *"the best strategy is to use a real-time quantitative polymerase chain reaction (qPCR) and immunofluorescence assay in*

tandem" (PMID: 31587667). - **Supportive labs:** mild thrombocytopenia, leukopenia/leukocytosis, elevated transaminases, elevated CRP (non-specific). - **Culture:** possible but hazardous/slow; restricted to reference labs.

Genetic testing / omics diagnostics. Human genetic testing is **not applicable**. Pathogen genomics (WGS/PCR) is used for organism identification and epidemiology, not for host diagnosis.

Clinical criteria & differential diagnosis. Because "*clinical features overlap, serologic tests are necessary to distinguish QTT from other endemic Australian rickettsial diseases (scrub and murine typhus)*" (PMID: 1962102). Key differentials: **scrub typhus** (*Orientia tsutsugamushi*), **murine typhus** (*R. typhi*), and **Flinders Island spotted fever** (*R. honei*), which "*extends beyond Flinders Island*" across south-east Australia (PMID: 16175900); also dengue, leptospirosis, and Q fever in the same region.

Screening. No population screening (acute, sporadic, environmentally acquired infection).

Section 11 — Outcome / Prognosis

- **Overall prognosis:** Good with prompt doxycycline; usually mild and self-limited (PMID: 1962102).
- **Severe disease / mortality:** A real minority. In the North Queensland audit, 3/37 (8%) SFG patients had severe disease (1 death, 2 permanent disability) versus 0/95 scrub typhus ($p = 0.02$) (PMID: 31318873); a 23-year ICU series included 9 QTT patients (median APACHE II 13) (PMID: 32959771).
- **Recovery:** Complete recovery is the norm with treatment; residual disability is rare.
- **Prognostic factors:** older age, ICU-level illness, and **delayed defervescence (>48 h)**, which is "*more common in patients with severe disease*" (PMID: 33075531); paediatric age predicts a benign course (PMID: 32252063).

Section 12 — Treatment

First-line pharmacotherapy — doxycycline (NCIT: C312; tetracycline-class antibiotic). Doxycycline is the standard of care. In the analogous Mediterranean spotted fever cohort, *"Doxycycline administration prior to deterioration of disease (in 31 patients) protected patients from development of severe MSF"* (RR 0.248, 95% CI 0.08–0.76) with earlier defervescence (3.0 vs 7.1 days) ([PMID: 21642652](#)).

Avoid fluoroquinolones. In the same analysis, *"fluoroquinolone treatment was associated with increased MSF disease severity"* (RR 2.53, 95% CI 1.40–4.55) — so fluoroquinolones should be avoided in SFG rickettsioses ([PMID: 21642652](#)).

Alternatives. Macrolides (e.g., azithromycin) are considered for children/pregnancy in SFG rickettsioses, though experimental data (canine RMSF) suggest azithromycin is *less* efficacious than doxycycline ([PMID: 10103185](#)); chloramphenicol is a historical alternative.

Supportive care. Antipyretics, fluids, and — for the severe/ICU subset — organ support (vasopressors, ventilation) as needed ([PMID: 32959771](#)).

Advanced/experimental therapeutics, pharmacogenomics, surgery, targeted/immuno/cell/gene therapy: Not applicable — QTT is cured by a short antibiotic course.

Treatment-response caveat. Persistent fever >48 h after starting doxycycline does **not** necessarily indicate misdiagnosis; delayed defervescence occurs in ~56% and correlates with severity ([PMID: 33075531](#)).

Suggested NCIT term: Doxycycline (NCIT:C312); Tetracycline antibiotic therapy.

Section 13 — Prevention

- **Primary prevention:** No vaccine exists. Prevention relies on **tick-bite avoidance** — protective clothing, DEET/permethrin repellents, avoiding tick habitat during peak season, systematic tick checks, and **prompt, correct tick removal** after exposure in endemic eastern Australia.
- **Chemoprophylaxis:** Routine post-tick-bite antibiotic prophylaxis is **not** recommended; management is watchful waiting with early empiric doxycycline if symptoms develop.
- **Secondary prevention:** Early recognition and prompt doxycycline to prevent progression to severe disease ([PMID: 21642652](#)).

- **Tertiary prevention:** Supportive/ICU care to prevent complications in severe cases.
- **Public health:** Clinician and public education (given rising ascertainment, [PMID: 31318873](#)), notifiable-disease surveillance, and environmental/personal tick-control measures. Genetic counseling is not applicable.

Section 14 — Other Species / Natural Disease

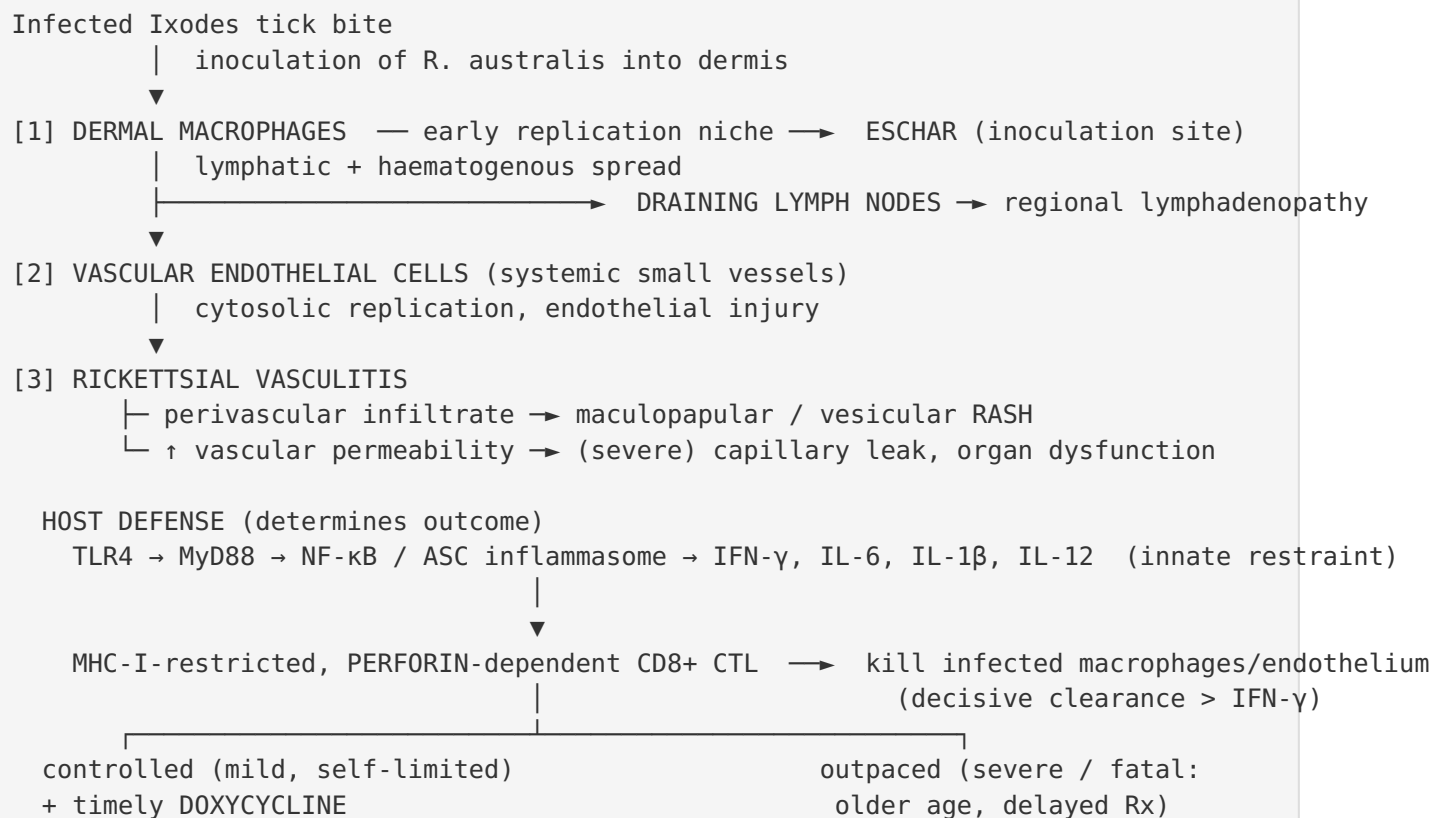
- **Vectors (NCBI Taxonomy):** *Ixodes holocyclus* (paralysis tick) and *Ixodes tasmani* are the only two confirmed vectors of *R. australis* ([PMID: 1962102](#)). *I. holocyclus* is "the cause of the neurotoxic disease tick paralysis in humans and mammals" in densely populated eastern Australia ([PMID: 25434042](#)).
- **Reservoir hosts:** native small mammals — rodents, bandicoots, and marsupials — maintain *R. australis* in nature; humans are accidental dead-end hosts.
- **Zoonotic transmission:** QTT is a **zoonosis** transmitted only via tick bite; there is no human-to-human transmission.
- **Comparative biology / cross-species susceptibility:** *R. australis* has been detected molecularly in *Ixodes ricinus* ticks in Europe, indicating a broader potential vector/host range than the Australian endemic cycle ([PMID: 18355299](#)); DNA identical/very close to *R. australis* has been found in mite endosymbiont surveys ([PMID: 31549736](#)). No significant naturally occurring clinical disease from *R. australis* is documented in companion animals (contrast *R. rickettsii*, which causes canine Rocky Mountain spotted fever, [PMID: 10103185](#)).

Section 15 — Model Organisms

- **Principal model — mouse (*Mus musculus*, NCBI Taxon 10090):** The C57BL/6 mouse is the workhorse for *R. australis* immunopathogenesis. Gene-knockout and adoptive-transfer studies established the protective hierarchy of host immunity:
- **Perforin^{-/-} and IFN-γ^{-/-} mice** are "more than 100-fold more susceptible" and **MHC class I^{-/-} mice** ">50,000-fold" more susceptible to lethal *R. australis*; "CTL activity was more critical to recovery from rickettsial infection than were the effects of IFN-gamma" ([PMID: 11179362](#)).
- **MyD88^{-/-} mice** show "severely impaired bacterial clearance in vivo" with blunted IFN-γ/IL-6/IL-1β and defective dendritic-cell maturation ([PMID: 26755162](#)).

- **TLR4/ASC-inflammasome** contributes to host immunity against *R. australis* (PMID: 32014896).
- Macrophage-tropism studies use *R. australis* in mice to show macrophages as an early target (PMID: 30297526).
- **Model type:** mammalian *in vivo* (genetic knockouts: *Prf1*, *Ifng*, *B2m*/MHC-I, *Myd88*, *Pycard*/ASC); plus *in vitro* infection of macrophage-like and microvascular endothelial cells.
- **Phenotype recapitulation:** The mouse model faithfully reproduces disseminated rickettsial infection, endothelial/macrophage targeting, and lethal severe disease, making it well suited to dissect protective immunity.
- **Limitations:** Mouse studies emphasize immunology and lethality; they do not reproduce the human eschar/rash phenotype in detail, and inbred knockouts do not capture human genetic diversity.

Mechanistic Model (synthesis)



Evidence Base

PMID	Title (abbrev.)	Role in this report
1962102	<i>Spotted fever group rickettsial infections in Australia</i>	Etiology, vectors, geography, mild phenotype, serologic diagnosis, differentials
22933759	<i>Genome sequence of R. australis</i>	1.29-Mb genome, 2 plasmids, <i>R. akari</i> similarity, 1950 isolation
11034486	<i>ompA/ompB of R. australis</i>	"Most divergent" SFG rickettsia; SFG classification
11491333	<i>Phylogeny via gene D</i>	Places <i>R. australis</i> in the <i>R. akari</i> group
9103608	<i>gltA citrate synthase phylogeny</i>	Confirms SFG/ <i>R. akari</i> -group placement
19327117	<i>Host-cell interactions with pathogenic Rickettsia</i>	Endothelial tropism → "rickettsial vasculitis" core mechanism
30297526	<i>Rickettsial macrophage tropism</i>	Macrophages as early target preceding endothelium
32014896	<i>ASC inflammasome / TLR4 vs R. australis</i>	Innate immune control of <i>R. australis</i>
26755162	<i>MyD88 in dendritic cells</i>	MyD88-dependent protective Th1 immunity; impaired clearance in KO
11179362	<i>Cytotoxic T lymphocytes in rickettsial clearance</i>	CD8/perforin/MHC-I decisive, > IFN- γ
31318873	<i>Rickettsial diseases in North Queensland</i>	Severity (8% severe SFG; 1 death), rising incidence
32959771	<i>ICU case series (QTT + scrub typhus)</i>	9 ICU QTT cases; APACHE II 13; presentation timing
32252063	<i>Rickettsial infection in children</i>	Benign paediatric course
33075531	<i>Prompt defervescence dogma</i>	56% delayed defervescence; links to severity
21642652	<i>Risk factors for malignant MSF</i>	Doxycycline protective; fluoroquinolones deleterious
31587667	<i>Diagnosis of SFG rickettsioses</i>	qPCR + IFA in tandem
22092999		

PMID	Title (abbrev.)	Role in this report
	<i>Real-time PCR for rickettsial diagnosis</i>	Validated <i>R. australis</i> qPCR; eschar/blood samples
31268225	<i>Tasmanian rickettsial hotspots</i>	Incidence 0.6/100,000/yr; seasonality, older males
25434042	<i>Ixodes holocyclus as vector</i>	<i>I. holocyclus</i> human-biting tick / paralysis
16175900	<i>Not only 'Flinders Island' spotted fever</i>	<i>R. honei</i> differential across SE Australia
10103185	<i>Doxycycline vs azithromycin vs trovafloxacin (canine RMSF)</i>	Comparative antibiotic efficacy
18355299	<i>Ixodes ricinus bacterial communities</i>	<i>R. australis</i> DNA detected in European ticks

Limitations and Knowledge Gaps

- Mechanistic data are largely from mouse models.** The pivotal immunology (perforin/CD8 CTL dominance, MyD88, TLR4/ASC) derives from *R. australis* mouse experiments ([PMID: 11179362](#); [PMID: 26755162](#); [PMID: 32014896](#)); human immunopathology is inferred, not directly measured.
- Endothelial-injury step is cross-group inference.** The "rickettsial vasculitis" mechanism is established for SFG rickettsiae broadly ([PMID: 19327117](#)) and applied to *R. australis*; species-specific human histopathology of QTT is sparsely quantified.
- Epidemiology conflates agents.** The best incidence figure (0.6/100,000/yr, Tasmania) largely reflects *R. honei* (Flinders Island spotted fever), not pure *R. australis* ([PMID: 31268225](#); [PMID: 16175900](#)). Mainland *R. australis* incidence is not precisely quantified.
- Treatment evidence borrows from MSF.** The strongest doxycycline-benefit / fluoroquinolone-harm data come from *R. conorii* MSF ([PMID: 21642652](#)); no QTT-specific randomized trial exists.
- No human host-genetics data.** Whether human innate-immunity polymorphisms modulate QTT severity is unknown.
- No quality-of-life or long-term-outcome studies** specific to QTT.

Proposed Follow-up Experiments / Actions

1. **Species-resolved surveillance:** Report *R. australis* versus *R. honei* separately in Australian notifiable-disease data to obtain a true QTT incidence and geographic map.
 2. **Prospective QTT cohort** capturing eschar/blood qPCR positivity, IFA kinetics, defervescence time, severity predictors, and QoL (EQ-5D/SF-36) at 30/90 days.
 3. **Human immunogenetics/immunophenotyping:** Test whether *TLR4/MYD88*/inflammasome/*PRF1* variants or peripheral CD8 CTL responses correlate with QTT severity, to validate the mouse-derived model in humans.
 4. **Diagnostic optimization:** Define the sensitivity/timing window of eschar-swab qPCR versus blood qPCR versus paired IFA in prospectively enrolled QTT patients ([PMID: 31587667](#); [PMID: 22092999](#)).
 5. **Treatment-timing study:** Quantify the effect of early empiric doxycycline on progression to severe QTT specifically (mirroring the MSF analysis), and formally test avoidance of fluoroquinolones.
 6. **Vector/reservoir ecology & One Health mapping** of *I. holocyclus*/*I. tasmani* infection prevalence and reservoir hosts to guide targeted public-health messaging in expanding peri-urban endemic zones ([PMID: 31318873](#)).
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Evidence-source key: Human clinical (case series/audits/serosurveys — PMIDs 1962102, 31318873, 32959771, 32252063, 33075531, 21642652, 31587667, 22092999, 16175900); Model organism (mouse — PMIDs 11179362, 26755162, 32014896, 30297526); Microbiology/genomics (PMIDs 22933759, 11034486, 11491333, 9103608); In vitro/vector (PMIDs 25434042, 18355299, 10103185).
